Cloud-Based Voting System

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OVERVIEW

Introduction & Objectives

Cloud Infrastructure

Web-based voting application eliminating physical polling constraints with cloud scalability.

Secure Authentication

Voter identity validation through OTP verification and encrypted credential verification.

Automated Results

Real-time automated vote processing and instant election result tallying.

Core Internship Goal

Practical implementation of full-stack web architecture backed by AWS cloud services and secure DB operations.

SYSTEM EVOLUTION

Existing vs. Proposed Solution

Traditional Voting System

- ✕ Physical polling stations required
- ✕ Heavy manual effort and paper consumption
- ✕ Time-consuming manual vote counting
- ✕ High operational costs & potential human error

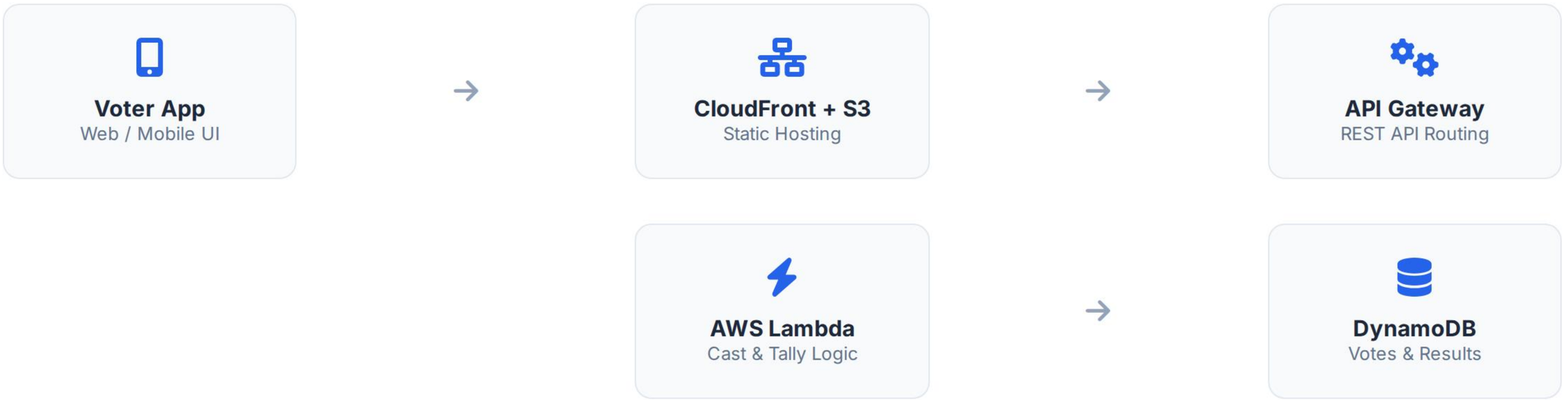
Cloud-Based Solution

- ✓ Remote electronic voting accessibility
- ✓ Centralized, encrypted cloud data storage
- ✓ Instant automated vote tallying
- ✓ Strict single-vote prevention checks

Key Advantage: High accessibility, lower administrative overhead, and zero physical infrastructure dependency.

SYSTEM DESIGN

AWS Cloud Architecture



CORE FUNCTIONALITIES

Proposed Methods & Capabilities

1. Registration & OTP

Voter identity validation via personal details and OTP verification.

2. Secure Vote Casting

Authenticated candidate selection with encrypted submission.

3. Duplicate Prevention

Strict status checks preventing voters from casting multiple ballots.

4. Cloud Storage

Centralized, scalable, and redundant database hosting.

5. Automatic Counting

Real-time automated vote processing minimizing manual error.

6. Admin Control

Centralized dashboard to manage candidates, voters, and results.

LOGIC FLOW

Core Execution Algorithms

Vote Casting Logic

1. Authenticate voter identity & OTP status.
2. Fetch active candidate election list.
3. Validate voter eligibility (`voted == false`).
4. Encrypt selected vote data payload.
5. Write record to Cloud DB & mark voted status.
6. Return confirmation receipt to voter.

Tallying Algorithm

1. Retrieve stored vote records from database.
2. Decrypt votes using server-side security key.
3. Aggregate total vote count per candidate.
4. Identify candidate with maximum valid votes.
5. Publish final results to Admin dashboard.
6. Generate system audit log summary.

CODE BASE

Frontend Implementation Highlights

// Sample Voting & Duplicate Check Logic

```
function vote() {
  if (!loginStatus)
    return "Please Login First";

  if (voted)
    return "You have already voted";

  let candidate = document
    .getElementById("candidate").value;

  votes[candidate]++;
  voted = true;

  document.getElementById("message")
    .innerHTML = "Vote Submitted Successfully";
}
```

Frontend Technologies

HTML5, CSS3, Vanilla JavaScript for client-side validation and responsive dashboard rendering.

Backend Runtime

Node.js with Express.js REST API endpoints managing server requests and session status.

APPLICATION INTERFACE

System Execution Screens

1. Landing Page
Home portal & role selection

2. Voter Auth
Voter ID & OTP login

3. Candidate Selection
Cast ballot securely

4. Result Dashboard
Live election analytics

5. Admin Portal
Manage voters & election lifecycle

EVALUATION

System Analysis

System Advantages

- ✓ **High Security:** Encryption & OTP authentication
- ✓ **Fast Processing:** Instant vote aggregation
- ✓ **Scalability:** AWS infrastructure handles traffic surges
- ✓ **Zero Paper Waste:** Eco-friendly digital execution

Limitations & Challenges

- ⚠ **Network Dependency:** Requires reliable internet connectivity
- ⚠ **Cyber Threats:** Requires advanced DDoS/threat protections
- ⚠ **Digital Divide:** Non-technical users need simple interfaces

SUMMARY & ROADMAP

Conclusion & Future Scope

Project Conclusion

The project successfully demonstrates how cloud computing and web technologies can transform traditional voting. Automated vote counting, OTP verification, and AWS integration ensure secure, rapid, and transparent election management.

Future Enhancements

- **Biometric Verification:** Facial/fingerprint matching
- **Blockchain Integration:** Immutable ledger audit trail
- **Native Mobile App:** iOS/Android app deployment
- **Multi-Factor Auth:** Advanced threat defense mechanisms

Thank You!